Computational Work 3

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Course: Computational Matrix Methods

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Abstract

This work investigates two least-squares problems using data derived from the student's enrollment number. First, an overdetermined linear system $A\mathbf{x} \approx \mathbf{b}$, with $A \in \mathbb{R}^{5\times 4}$, is solved using the QR factorization $A = QR$. The least-squares solution is obtained by solving the resulting upper triangular system. The existence of a unique solution is discussed in terms of the rank of the coefficient matrix.

Second, a quadratic polynomial is fitted to six experimental data points using the least-squares formulation. The associated design matrix is constructed, and the polynomial coefficients are determined computationally. The quality of the fit is assessed by plotting the experimental data and the fitted curve and calculating the mean squared error.

The numerical implementation explicitly constructs the principal linear algebra algorithms, avoiding high-level routines for QR factorization and least-squares solutions. This approach connects the mathematical foundations of least squares with their computational implementation and numerical evaluation.

Keywords: QR factorization; least squares; quadratic polynomial fitting; numerical linear algebra.

1 Problem Formulation

This work addresses two computational problems involving least-squares methods. The numerical data are constructed from the eight digits of the student's enrollment number. Any zero digit is replaced by one before constructing the matrices and vectors.

1.1 Problem 1: QR Factorization and Linear Least Squares

Consider the overdetermined linear system

$$A\mathbf{x} \approx \mathbf{b}, \tag{1}$$

where $A \in \mathbb{R}^{5\times 4}$, $\mathbf{x} \in \mathbb{R}^4$, and $\mathbf{b} \in \mathbb{R}^5$. The coefficient matrix and right-hand-side vector are defined by

$$A = \begin{bmatrix} m_1+2 & m_2 & -m_3 & m_4 \\ m_2+1 & -m_3 & m_4+2 & m_5 \\ m_3+2 & -m_4 & m_5+1 & -m_6 \\ -m_4 & m_5+2 & m_6 & m_7 \\ -m_5 & m_6+1 & m_7+2 & m_8 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} m_1+m_2 \\ -m_3 \\ m_4+m_5 \\ -m_6 \\ m_7+m_8 \end{bmatrix}. \tag{2}$$

The first objective is to compute the reduced QR factorization

$$A = QR, \tag{3}$$

where $Q \in \mathbb{R}^{5\times 4}$ has orthonormal columns and $R \in \mathbb{R}^{4\times 4}$ is upper triangular.

The least-squares solution is defined as

$$\mathbf{x}^* = \arg\min_{\mathbf{x}\in\mathbb{R}^4} \|A\mathbf{x} - \mathbf{b}\|_2^2. \tag{4}$$

The uniqueness of this solution depends on the column rank of A. In particular, the solution is unique if and only if $\text{rank}(A) = 4$. This condition will be assessed computationally for the given enrollment number.

1.2 Problem 2: Quadratic Least-Squares Fitting

Consider six experimental data points (x_i, y_i), where

$$x_i = i, \qquad y_i = m_i \bmod 5, \qquad i = 1, \ldots, 6. \tag{5}$$

The objective is to approximate these data using a quadratic polynomial

$$p(x) = ax^2 + bx + c. \tag{6}$$

Defining the coefficient vector $\boldsymbol{\theta} = [a, b, c]^T$, the fitting problem can be expressed as

$$X\boldsymbol{\theta} \approx \mathbf{y}, \tag{7}$$

where

$$X = \begin{bmatrix} 1^2 & 1 & 1 \\ 2^2 & 2 & 1 \\ 3^2 & 3 & 1 \\ 4^2 & 4 & 1 \\ 5^2 & 5 & 1 \\ 6^2 & 6 & 1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{bmatrix}. \tag{8}$$

The polynomial coefficients are obtained by solving

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^3} \|X\boldsymbol{\theta} - \mathbf{y}\|_2^2. \tag{9}$$

Since the six values of x_i are distinct, the three columns of X are linearly independent. Therefore, $\text{rank}(X) = 3$, and the least-squares solution is unique.

Finally, the quality of the fitted polynomial is evaluated by plotting the experimental points and

the fitted curve and by computing the mean squared error (MSE):

$$\text{MSE} = \frac{1}{6}\sum_{i=1}^{6}\left[y_i - p(x_i)\right]^2 . \tag{10}$$

References

[1] Golub, G. H., & Van Loan, C. F. (2013). *Matrix Computations* (4th ed.). Johns Hopkins University Press.

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